

Capture The Flag

Mission: Learn how Angular applications are built using reusable components.

Objective: Transform your first Angular application into a modular application using reusable components while following a basic Git workflow.

Flag 0 – Return to Base

Mission

Open the Angular project you created during the previous activity.

Tasks

- Open **Visual Studio Code**.
- Open your Angular project folder.
- Open the integrated terminal.
- Verify that the project still runs.
 - ng serve

Open:

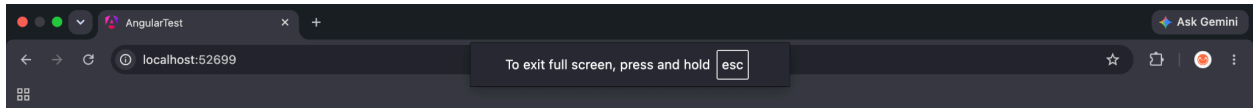
`http://localhost:4200`

If your Angular welcome page appears, proceed.



Evidence

- VS Code with project opened
- Terminal showing `ng serve`
- Browser displaying the Angular application



Flag 1 – Mission Branch

Mission

Every new feature must have its own Git branch.

Create a new branch named:

feature/components

Commands:



Evidence

- Terminal showing the successful branch creation
- Output of `git branch`

```
woj@2 ITFE403A_RIZAL % git checkout -b features/components
Switched to a new branch 'features/components'
woj@2 ITFE403A_RIZAL % git push -u origin features/components
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL.git
remote:
remote: Create a pull request for 'features/components' on GitHub by visiting:
remote:   https://github.com/W0jkurt/ITE403A_RIZAL/pull/new/features/components
remote:
To https://github.com/W0jkurt/ITE403A_RIZAL.git
 * [new branch]      features/components -> features/components
branch 'features/components' set up to track 'origin/features/components'.
woj@2 ITFE403A_RIZAL %
```

Flag 2 – Explore the Base

Mission

Investigate your Angular project before making changes.

Locate the following:

- app.html
- app.ts
- app.css
- app.spec.ts

Understand that these files belong to the **App Component**, the root component of the application.



Evidence

- Explorer showing the files
- `app.html` opened

```
TS app.config.ts
{ } app.css
</> app.html
TS app.routes.ts
TS app.spec.ts
TS app.ts
</> index.html
TS main.ts
{ } styles.css
```

Flag 3 – Clear the Battlefield

Mission

Remove the default Angular welcome page.

Delete everything inside:

app.component.html

Replace it with:

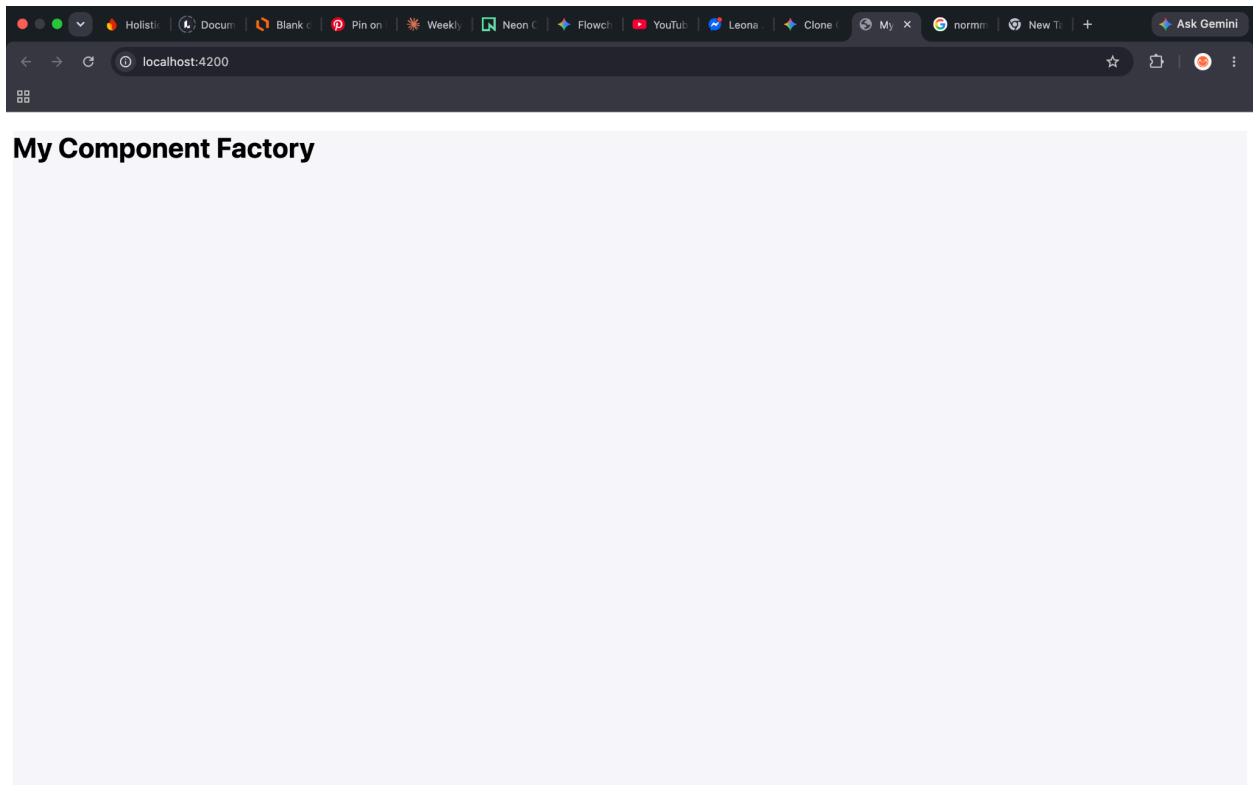
```
<h1>My Component Factory</h1>
```

Verify the browser updates.



Evidence

- Browser showing only the heading
- HTML file



Flag 4 – Build the Headquarters

Mission

Create a reusable Header component.

Generate:

```
ng generate component components/header
```

Angular should automatically generate:

```
header.ts
header.html
header.css
header.spec.ts
```



Evidence

- Terminal

- Explorer showing generated files

Flag 5 – Raise the Banner

Mission

Inside the Header component, create a simple header.

Example:

```
<header>  
  <h2>Don Bosco Student Portal</h2>  
</header>
```

Use the component inside:

```
app.html  
<app-header></app-header>
```

Verify it appears in the browser.

Evidence

- Browser
- HTML code

```
woj@2 angular test % ls
angular-test
woj@2 angular test % cd angular-test
woj@2 angular-test % ng generate component components/header
CREATE src/app/components/header/header.css (0 bytes)
CREATE src/app/components/header/header.spec.ts (531 bytes)
CREATE src/app/components/header/header.ts (186 bytes)
CREATE src/app/components/header/header.html (21 bytes)
woj@2 angular-test %
```

Flag 6 – Secure the Exit

Mission

Create a Footer component.

Use the same command previously, but now for the footer.

Display:

<footer>

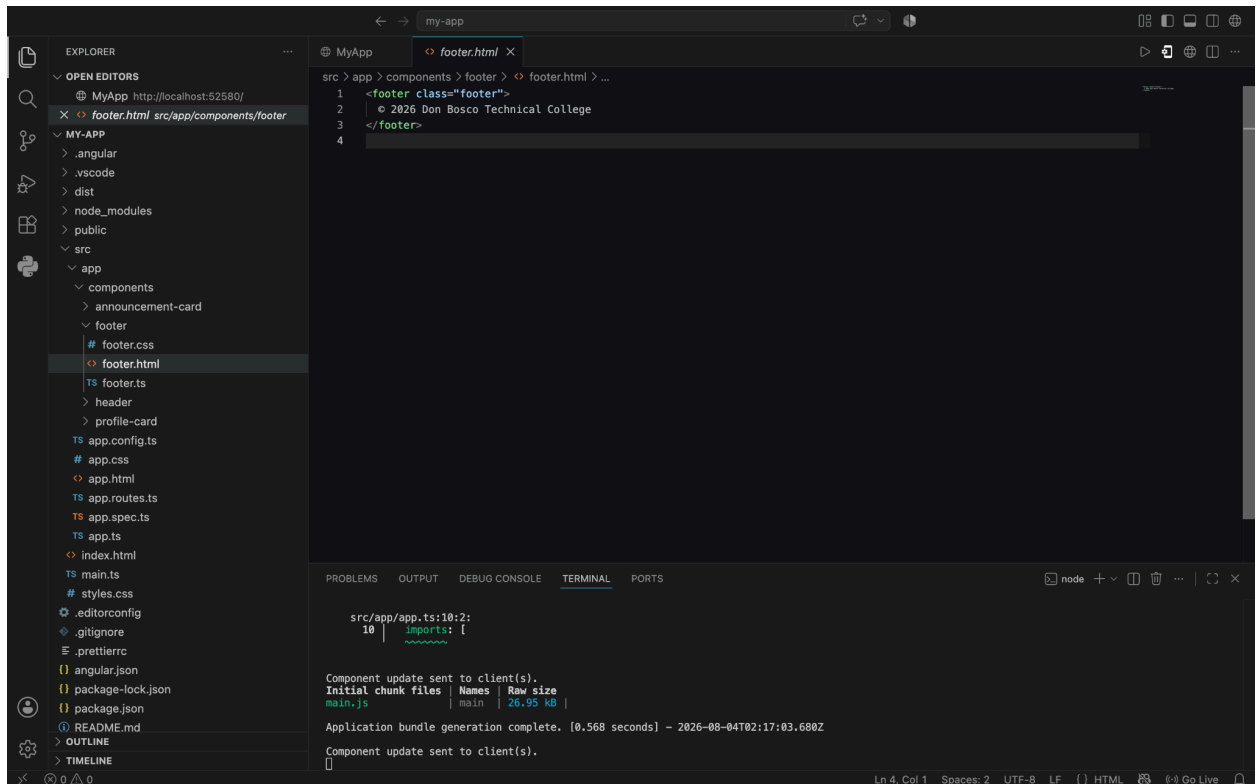
© 2026 Don Bosco Technical College

</footer>

Insert it into the App Component.

Evidence

- Browser
- Generated files



Flag 7 – Create the Profile Card

Mission

Generate a reusable Profile Card.

Use the command to create a new component called profile-card.

Create a card containing:

- Profile Picture
- Name

- Course
- Year Level

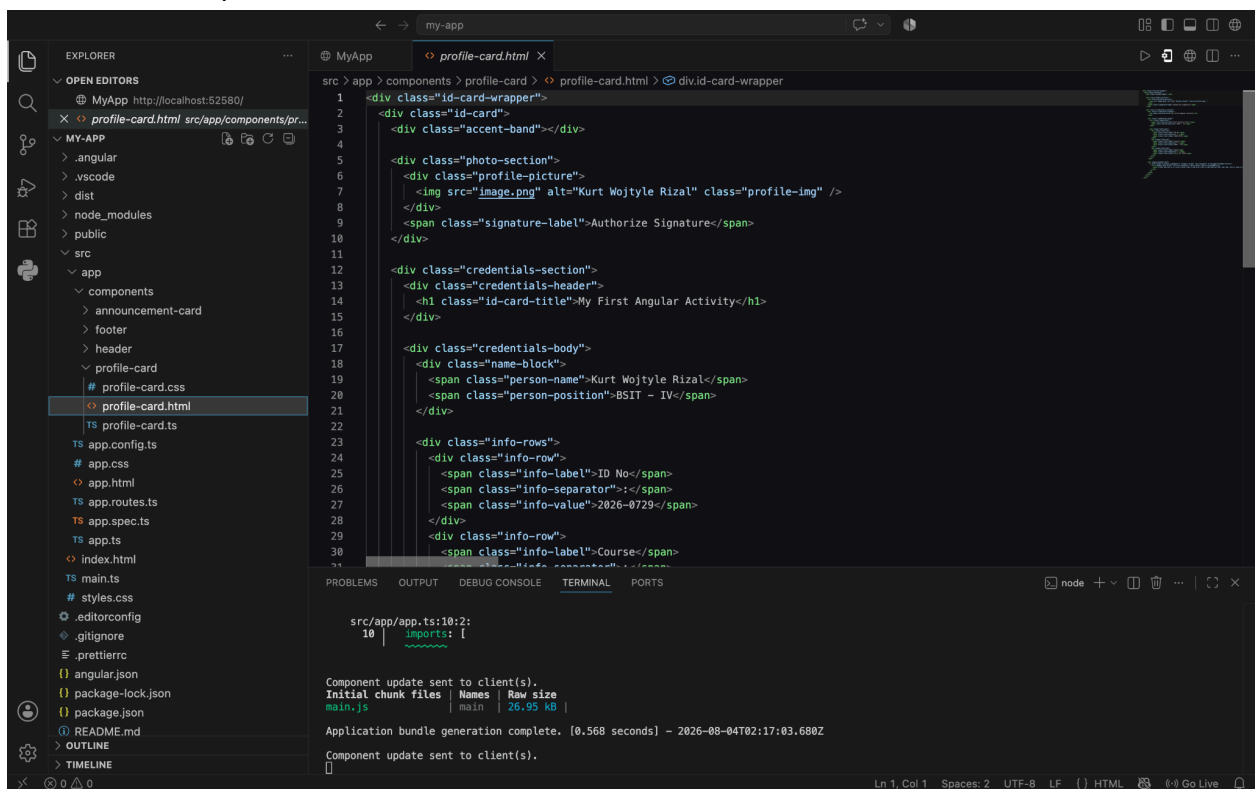
(Design is optional)

Insert it into:

app.html

Evidence

- Browser
- Profile Card component



The screenshot shows a VS Code editor with the file explorer on the left, the editor window in the center, and the terminal at the bottom. The file explorer shows the project structure with the following files and folders:

- MyApp
 - profile-card.html
 - src
 - app
 - components
 - announcement-card
 - footer
 - header
 - profile-card
 - profile-card.css
 - profile-card.html
 - profile-card.ts
 - app.config.ts
 - app.css
 - app.html
 - app.routes.ts
 - app.spec.ts
 - app.ts
 - index.html
 - main.ts
 - styles.css
 - package-lock.json
 - package.json
 - README.md
 - OUTLINE
 - TIMELINE

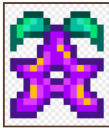
The editor window shows the content of profile-card.html:

```
1 <div class="id-card-wrapper">
2 <div class="id-card">
3 <div class="accent-band"></div>
4
5 <div class="photo-section">
6 <div class="profile-picture">
7 
8 </div>
9 <span class="signature-label">Authorize Signature</span>
10 </div>
11
12 <div class="credentials-section">
13 <div class="credentials-header">
14 <h1 class="id-card-title">My First Angular Activity</h1>
15 </div>
16
17 <div class="credentials-body">
18 <div class="name-block">
19 <span class="person-name">Kurt Wojtyle Rizal</span>
20 <span class="person-position">BSIT - IV</span>
21 </div>
22
23 <div class="info-rows">
24 <div class="info-row">
25 <span class="info-label">ID No</span>
26 <span class="info-separator"></span>
27 <span class="info-value">2026-0729</span>
28 </div>
29 <div class="info-row">
30 <span class="info-label">Course</span>
31 <span class="info-value">BSIT - IV</span>
32 </div>
33 </div>
34 </div>
35 </div>
36 </div>
```

The terminal output shows the following messages:

```
src/app/app.ts:10:2:
10 | imports: [
   | ~~~~~
Component update sent to client(s).
Initial chunk files | Names | Raw size
main.js | main | 26.95 KB |
Application bundle generation complete. [0.568 seconds] - 2026-08-04T02:17:03.680Z
Component update sent to client(s).
```

Don Bosco Student Portal



MY FIRST ANGULAR ACTIVITY

Kurt Wojtyle Rizal
BSIT - IV

ID No : 2026-0729
Course : BSIT - IV
Date : July 29, 2026

Authorize Signature



Flag 8 – Produce Announcement Cards

Mission

Generate one reusable Announcement Card.

Use the same command, creating the component announcement-card.

Design a card containing:

- Announcement Title
- Description
- Date

Do **NOT** create three separate components.

Instead, reuse the same component.

Example:

```
<app-announcement-card></app-announcement-card>
```

```
<app-announcement-card></app-announcement-card>
```

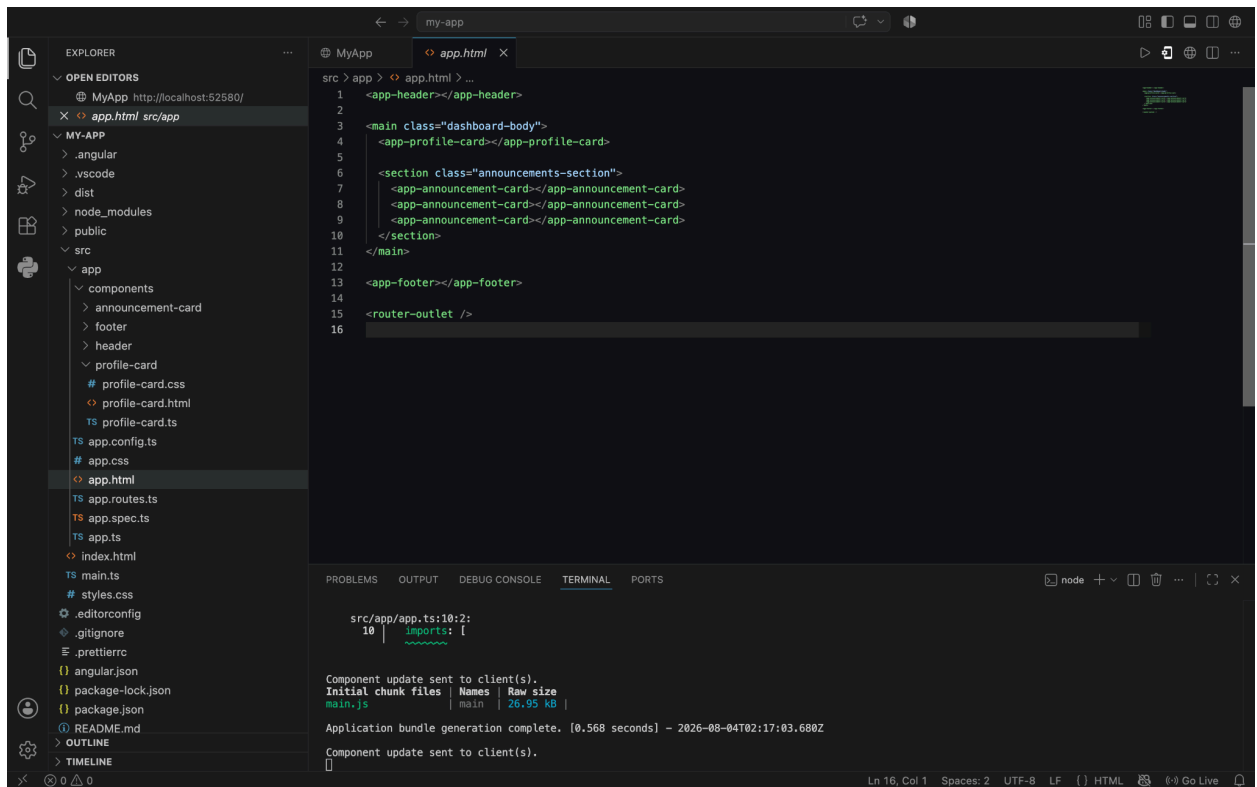
```
<app-announcement-card></app-announcement-card>
```



Evidence

- Browser

- App component showing three announcement cards



Flag 9 – Assemble the Dashboard

Your `app.html` should now act only as the page assembler.

Assemble your page to have this structure:

header

profile-card

announcement-card

announcement-card

announcement-card

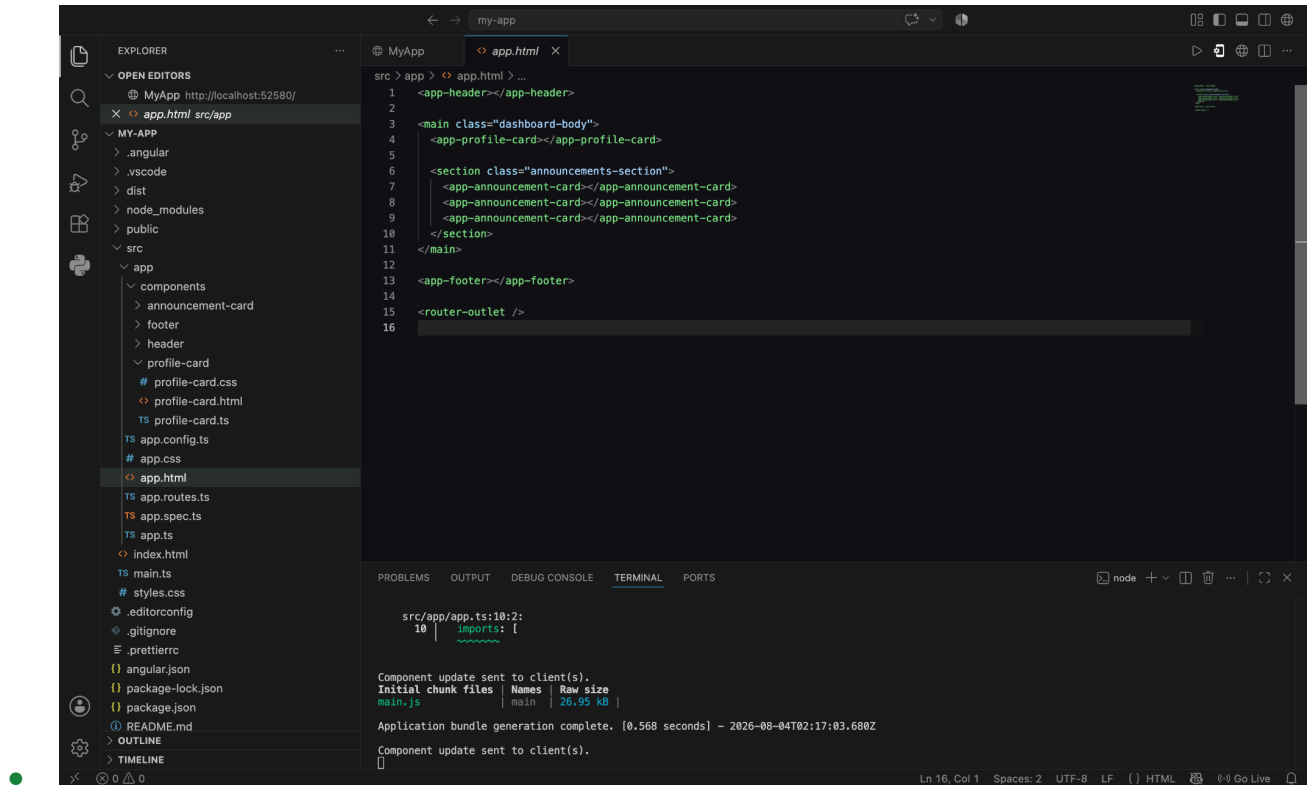
footer

Notice that almost no actual page content exists here. Only reusable components.



Evidence

- Browser
- App.html



Flag 10 – Client Changed Their Mind

Mission

The client sends an urgent request:

"Every announcement card should now have:

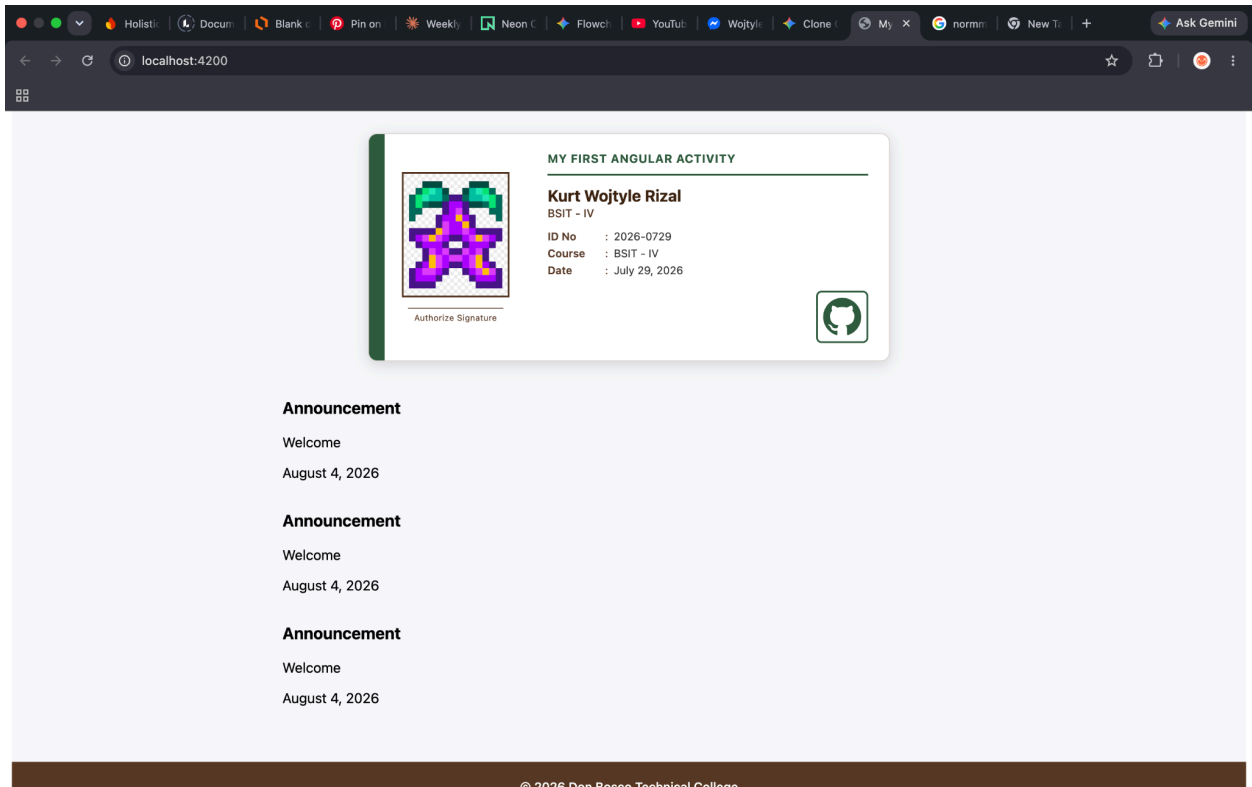
- Rounded corners
- A blue border
- Padding
- A subtle shadow"

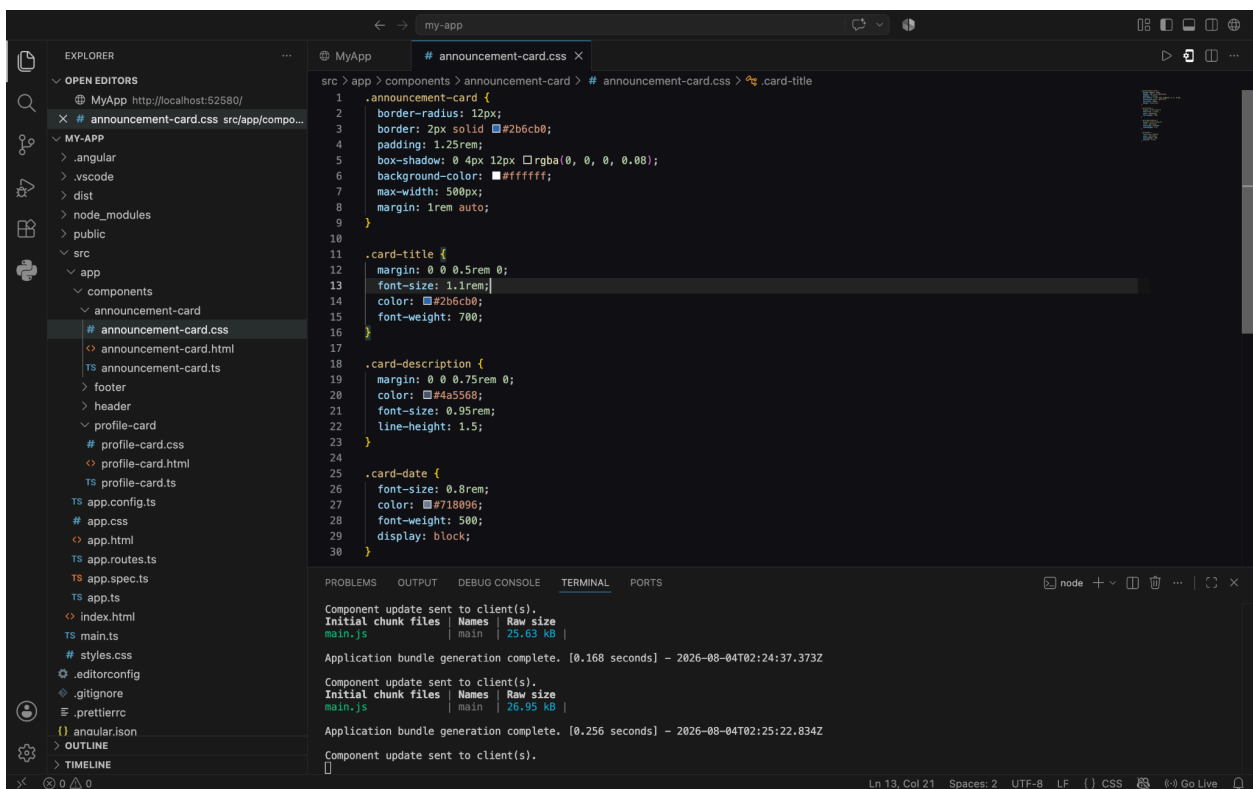
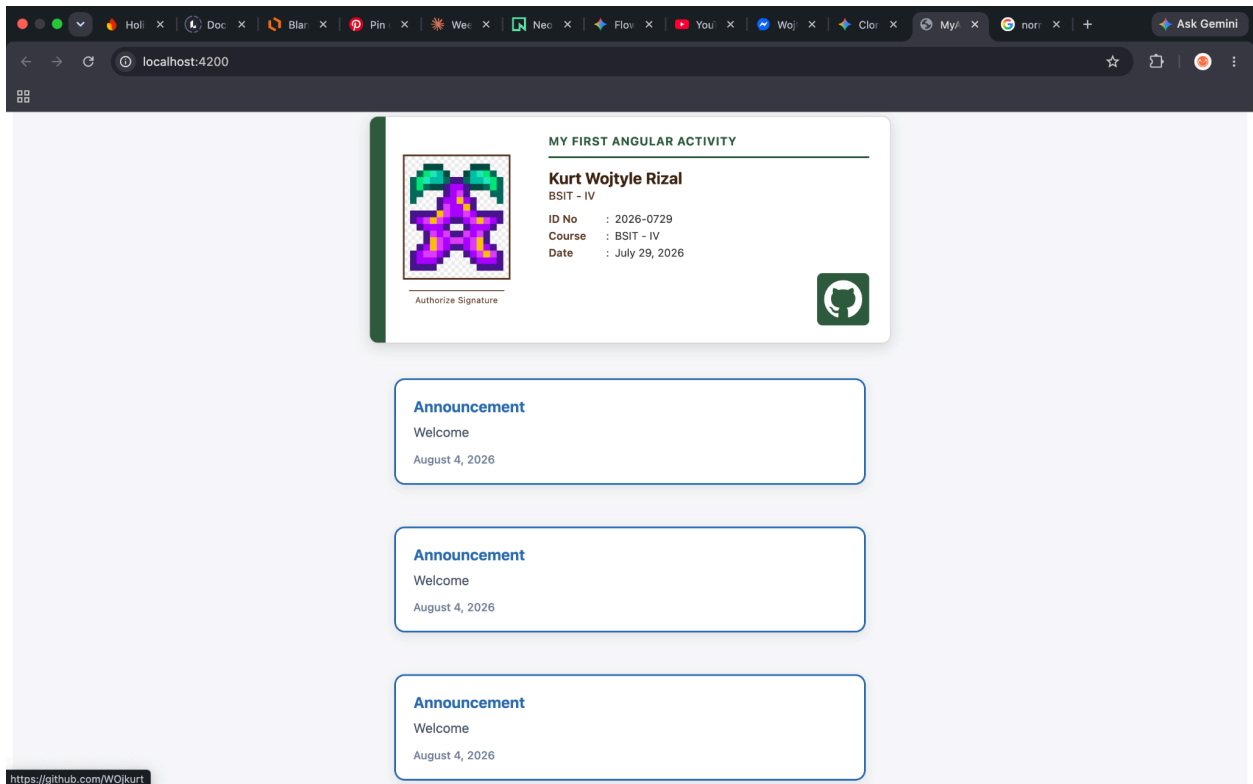
You must update **only one file**.

Apply your styles there and verify that **all announcement cards** update automatically.

Evidence

- Browser before and after the CSS change
- `Announcement-card.css` code





Flag 11 – Pushing Changes

Commit your completed work, then push to your repository.

Evidence

- Terminal commands for committing + pushing
- Screenshot of the Github Repository reflecting the pushed changes

```
woj@2 ITFE403A_RIZAL % git add .
woj@2 ITFE403A_RIZAL % git commit -m "Act 2"
On branch feature/components
Your branch is up to date with 'origin/feature/components'.

nothing to commit, working tree clean
woj@2 ITFE403A_RIZAL % git add .
woj@2 ITFE403A_RIZAL % git commit -m "Act 2"
[feature/components 11a4839] Act 2
 2 files changed, 2 insertions(+), 2 deletions(-)
woj@2 ITFE403A_RIZAL % git push origin
Enumerating objects: 19, done.
Counting objects: 100% (19/19), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 881 bytes | 440.00 KiB/s, done.
Total 10 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), completed with 6 local objects.
```